

GNN export and error-handling philosophy

GNN export (Generalized Notation Notation)

Exports package the program graph and compiled state-space model into the Active Inference Institute's **Generalized Notation Notation** plus a small set of interop targets:

- ▶ **GNN Markdown** (`model.gnn.md`) — canonical, human-readable Generalized Notation Notation organized into the section order defined in `../cogant/docs/export/README.md` (Model Metadata, Repository Metadata, Source Coverage, State Space, Observation Modalities, Actions/Policies, Connections, Factors, Transition Structure, Likelihood Structure, Preferences/Constraints, Time Settings, Parameterization, Ontology Mapping, Provenance, Confidence Scores, Rendering Hints, Validation Notes).
- ▶ **GNN JSON** (`model.gnn.json`) — machine-readable equivalent of the markdown sections, plus companion JSON files per section (`state_space.json`, `observations.json`, `actions.json`, `transitions.json`, and so on).
- ▶ **Interop exports** — GraphML and Parquet for analysis in

Error handling philosophy

The implementation distinguishes fatal configuration or parse failures from per-file errors that allow partial results. Validation aggregates warnings and inconsistencies into a report that travels with the bundle. This mirrors the layered error categories documented in the architecture (fatal, error, warning, info).